

What if the soundboard were the only thing that mattered?

GPU-Accelerated Soundboard Design with Physical Fabrication and Acoustic Verification

Co-Development Proposal: Industry + University

PIANOID — Real-Time GPU-Accelerated Physical Modeling Piano Synthesizer

The Problem

Why do some pianos sing while others stay dull — and why can't we answer this systematically?

Systematic soundboard design is economically and methodologically impossible with current tools. A concert grand has 230 strings coupled through a soundboard with thousands of resonance modes. Evaluating a single design change requires weeks of skilled labor and thousands in materials.

- "Killer octave" — poor sustain in 5th–6th octave. Mechanism understood; the **design solution is not**
- Design knowledge exists as **intuition accumulated over decades**, not as transferable parameters

“**Research question:** Can a closed-loop methodology — FEA design, real-time acoustic preview, physical fabrication, and quantified modal verification — make soundboard design systematic and reproducible?”

The core experiment: A/B synthesis — FEA-predicted modes vs. physically measured modes on the same engine, isolating model accuracy from synthesizer fidelity.

Research Landscape

Group	Focus	Key Contribution
Chabassier, Chaigne (Inria/ENSTA)	Full piano FE simulation	Energy-preserving string + soundboard + air (JASA 2013)
MAESSTRO (ANR consortium)	Soundboard CAD tool	FEA modal synthesis; string-soundboard coupling; offline (2022)
Ege, Boutillon (Polytechnique)	Experimental modal analysis	High-res soundboard characterization at mfg stages (2018)
Välimäki, Bank (Aalto/Budapest)	Real-time piano models	Efficient digital waveguide / IIR with psychoacoustic tuning
Modartt Pianoteq	Commercial physical model	~1000 modes/note, proprietary, no design pipeline
Weinreich (U. Michigan)	String coupling theory	Coupled polarization and double decay (JASA 1977)

All prior work operates within the conventional piano's constraints. No existing project treats the soundboard as an independent design variable, and no work closes the loop from FEA-derived modes through real-time synthesis to quantified physical verification.

What Already Exists: The Pianoid Platform

Working System — demonstrated at NAMM Show and in live concert performances

- **Synthesis algorithm** — full string set with waveguide model, 256 soundboard modes coupled to all strings, real-time on NVIDIA 4070 Ti with 30–50% margin
- **String-mode coupling network** — each soundboard mode is energised by string motion and returns energy back, creating continuous bidirectional exchange. Sympathetic resonance, pedal effects emerge naturally
- **Patented excitation system** — 8-channel electromechanical actuators, tested on a real piano soundboard
- **Modal measurement pipeline** — multi-band ESPRIT extraction with automated mode tracking across 78+ positions

What This Proposal Builds

- **FEA import pipeline** — extract string-mode coupling matrices from FEA eigenmode output
- **Algorithm improvement** — 256 → 2048 modes (in pipeline), string discretisation, hammer model
- **Soundboard fabrication pipeline** — rapid prototyping, material testing, encasing design
- **Perceptual validation** — ABX (forced-choice perceptual comparison) listening tests

Piano Modelling State of the Art — And Its Gaps

Capability	MAESSTRO	Pianoteq	Pianoid (current)	Proposed System
Synthesis speed	16 min/note	Real-time	Real-time, 48 kHz	Real-time, 48 kHz
Polyphony	Single note	Full	Full set of strings	Full set of strings
Multi-string coupling	No	Proprietary	256 modes	2048 modes
Soundboard from FEA	Yes	Hand-tuned	Measured only	FEA-derived
Modal measurement	No	No	ESPRIT pipeline	ESPRIT pipeline
Design verification	No	No	Not yet	A/B synthesis
Unconstrained soundboard	Conventional	Conventional	8-ch prototype	Multi-point excitation

Blue = proposed capabilities that this project delivers.

How We Enable the Loop

Closing the design loop requires two capabilities that no existing tool provides:

Real-Time Polyphonic Preview

GPU modal synthesis with full string-mode coupling lets a pianist **play** the designed soundboard — not listen to single offline notes. Design changes are heard instantly. Iteration takes minutes, not weeks.

Patented Excitation System

Electromechanical actuators replace physical strings entirely. No permanent tension, no iron plate, no frame. Soundboards are **swappable** — free mounting with no stress. A new design can be tested acoustically in hours.

Together, these make the FEA → preview → fabricate → measure → verify cycle practically executable.

Freeing the soundboard from its structural role also opens the door to designs that no conventional piano could accommodate.

The Closed Loop

FEA Design → GPU Preview → Fabrication → Measurement → A/B Evaluation

← ← ← *Below threshold? Revise FEA model, iterate from Phase 1* ← ← ←

Loop closure: quantified modal correlation (frequency, MAC, damping) defines go/no-go. A/B perceptual test validates convergence.

“ **Phases 1–2: Design & Preview** — FEA eigenvalue analysis. Per mode: **frequency, quality factor, modal mass, mode shape amplitudes at excitation positions**. Mode shapes define the string-mode coupling matrices. ”

“ **Phase 3: Fabrication** — Reduced-scale test boards. Spruce, plywood, composites. Carbon fiber ribs are **a hypothesis to test**, not a proven approach. ”

“ **Phase 4: Modal Verification** — Multi-channel ESPRIT + automated mode tracking. FEA-vs-measured: frequency, MAC (Modal Assurance Criterion), damping. ”

“ **Phase 5: A/B Synthesis & Evaluation** — **The core experiment:** same engine, FEA modes vs. measured modes. Isolates model accuracy from synthesizer fidelity. ABX listening tests, expert musician assessment. ”

Co-Development: Industry + University

(partner institution TBD)

Pianoid (Industry Partner) Brings

- **GPU synthesis engine** — real-time modal synthesis with string-mode coupling (working)
- **Patented excitation system** — 8-channel actuator prototype
- **Modal measurement pipeline** — ESPRIT extraction, automated mode tracking
- **FEA import tool development** — FEA eigenmode output → engine parameters
- **Engine and algorithm R&D** — mode capacity, string discretisation, hammer model

University Partner Brings

- **FEA expertise** — orthotropic material characterization, rib/bridge geometry, mesh convergence
- **Music perception science** — ABX test design, psychoacoustic evaluation, statistical analysis
- **Prototyping and testing facilities** — vibration lab, anechoic chamber, accelerometers, CNC
- **Acoustic radiation modelling** — sound field prediction; potential coupled structural-acoustic FEA
- **Credentialing in acoustics community** — peer-reviewed venues, conference networks

IP: commercial technology retained by Pianoid. Academic publications and modal datasets fully open.

Terms negotiated jointly

Why Now?

Technology Convergence

- **GPU compute is crossing the threshold** — real-time synthesis with waveguide strings and full string-mode coupling is feasible for the first time
- **MAESSTRO laid the FEA groundwork** — soundboard modal synthesis is proven; missing piece is real-time performance and a verification loop
- **Modal analysis tools are mature** — ESPRIT and related methods extract hundreds of modes reliably

Platform Readiness

- **Pianoid is demonstrated** — patented excitation, engine, and measurement pipeline shown at NAMM and in live concert performances
- **The unconstrained design concept is now executable** — the required combination of technologies has only recently become available

Growing Demand

- **Sustainable materials** — new candidate soundboard materials need acoustic evaluation tools
- **Digital twin methodology** — established in aerospace/automotive; extending to acoustic instruments is timely

Timeline and Risks

Phase	Months	Deliverable
Mode capacity scaling	1–2	2048-mode validation
String & hammer improvements	1–3	Upgraded waveguide + excitation
FEA import tool	1–3	Working pipeline
FEA soundboard design	2–5	Modal basis, 2+ designs
GPU preview & iteration	4–6	Playable virtual instruments
Prototype fabrication	4–8	Reduced-scale test boards
Measurement & verification	7–9	FEA-vs-measured data
A/B evaluation & performance	8–11	Playable prototype, ABX
Publication	10–12	Papers, dataset

“ **Algorithm improvements fail** — Mode scaling, discretisation, or hammer model exceed budget. **Fallback:** current algorithm stays as-is — proven in live performances.

Poor FEA correlation — Start simple. Discrepancies inform refinement — **this is the research.**

Prototype validity — Spruce baseline alongside experimental materials.

Partnership timeline — Parallel workstreams; industry work proceeds independently months 1–3.

”

Expected Outcomes

Scientific

- **First systematic study** of soundboard acoustics freed from string tension, case geometry, and fixed bridge constraints
- **A/B synthesis results** — controlled comparison of FEA vs. measured modes on the same engine
- **Design guidelines** — quantified sensitivity of acoustic features to structural parameters
- **Open modal dataset** — published characterizations at successive construction stages

Technological

- **Playable prototype** — fabricated soundboard with excitation system, live acoustic performance
- **Software pipeline** — integrated FEA-to-Pianoid import workflow
- **Modal measurement toolkit** — automated ESPRIT + tracking for instrument characterization

Industrial

- **Computer-aided soundboard design** with real-time preview
- **Sustainable materials evaluation** pathway
- Foundation for **novel acoustic instrument design** — soundboards unconstrained by piano geometry

What does a soundboard sound like when it is designed purely for sound?

Arbitrary geometry. Any material. Excitation at any point.
No tension. No frame. No constraints inherited from strings.
Every design decision driven by acoustic goals alone.

Now we have the tools to find out — and to design instruments that could never have existed before.

PIANOID